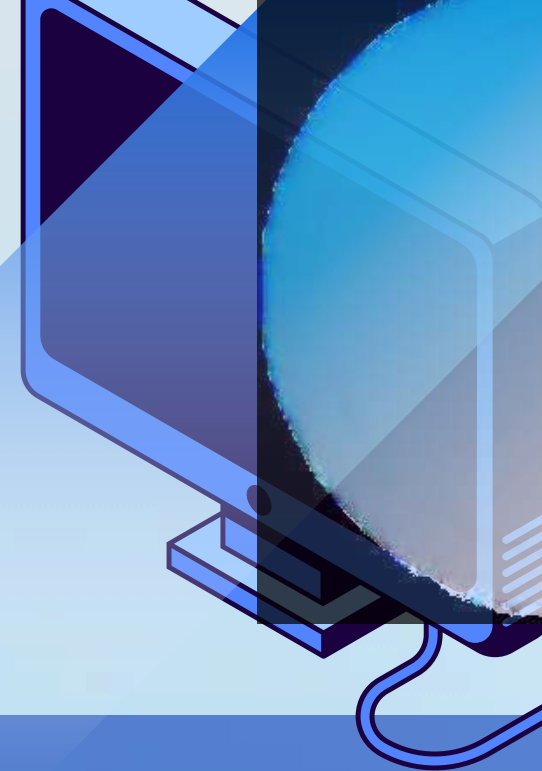




Klasifikasi Tweet Bencana

Menggunakan Regresi Logistik dan Naive Bayes



Masalah

- Twitter menjadi salah satu medsos dalam penyebaran informasi saat bencana terjadi.
- Lewat tweet, masyarakat dapat melaporkan situasi secara langsung dan real-time.
- Namun, tidak semua tweet yang memuat kata seperti “ablaze” benar-benar menggambarkan bencana; bisa saja hanya kiasan atau ungkapan metaforis.
 - “SOOOO PUMPED FOR ABLAZE ???? @southridgelifе”
 - “Man wife get six years jail for setting ablaze niece <http://t.co/eV1ahOUCZA>”



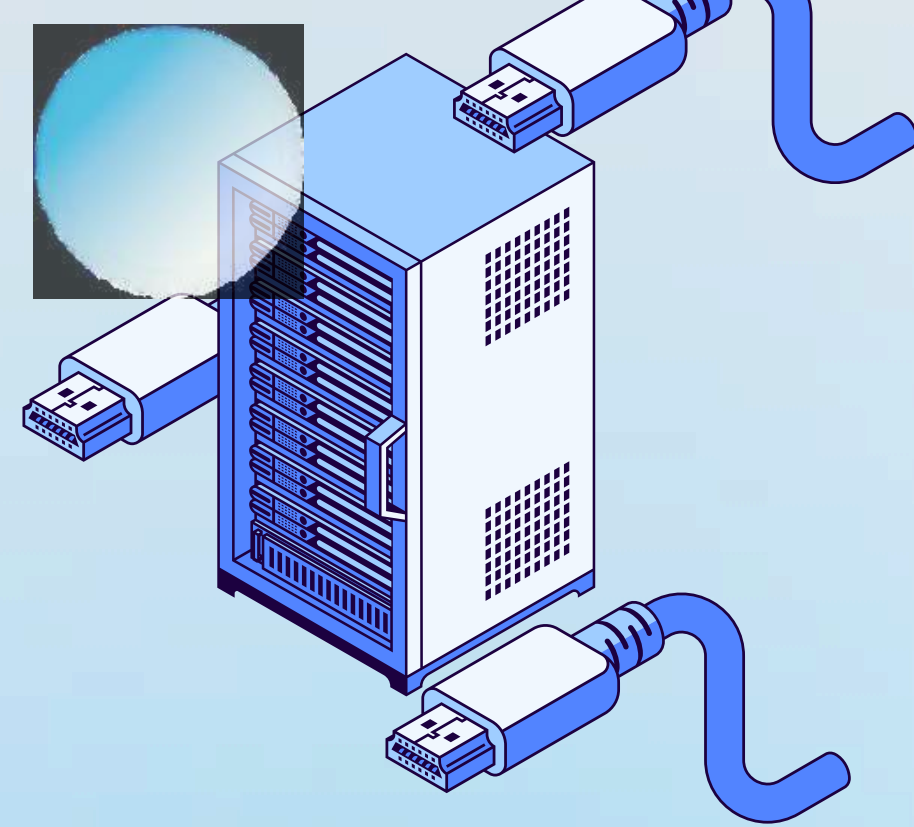
Tujuan

- Mengklasifikasikan tweet ke dalam kategori Bencana dan Non-bencana
- Membandingkan hasil klasifikasi Logistic Regression dan Naive Bayes
- Mengoptimalkan performa model
- Mendeteksi lokasi bencana



Dataset

- Source: Kaggle
- Train Data: 7.613 tweets.
- Main features:
 - text: Isi teks dari tweet.
 - keyword: Kata kunci tertentu dari tweet (bisa kosong).
 - location: Lokasi tempat tweet dikirim (bisa kosong).
 - target: Label ini menandakan apakah sebuah tweet tentang bencana nyata (1) atau bukan (0).





Exploratory Data Analysis

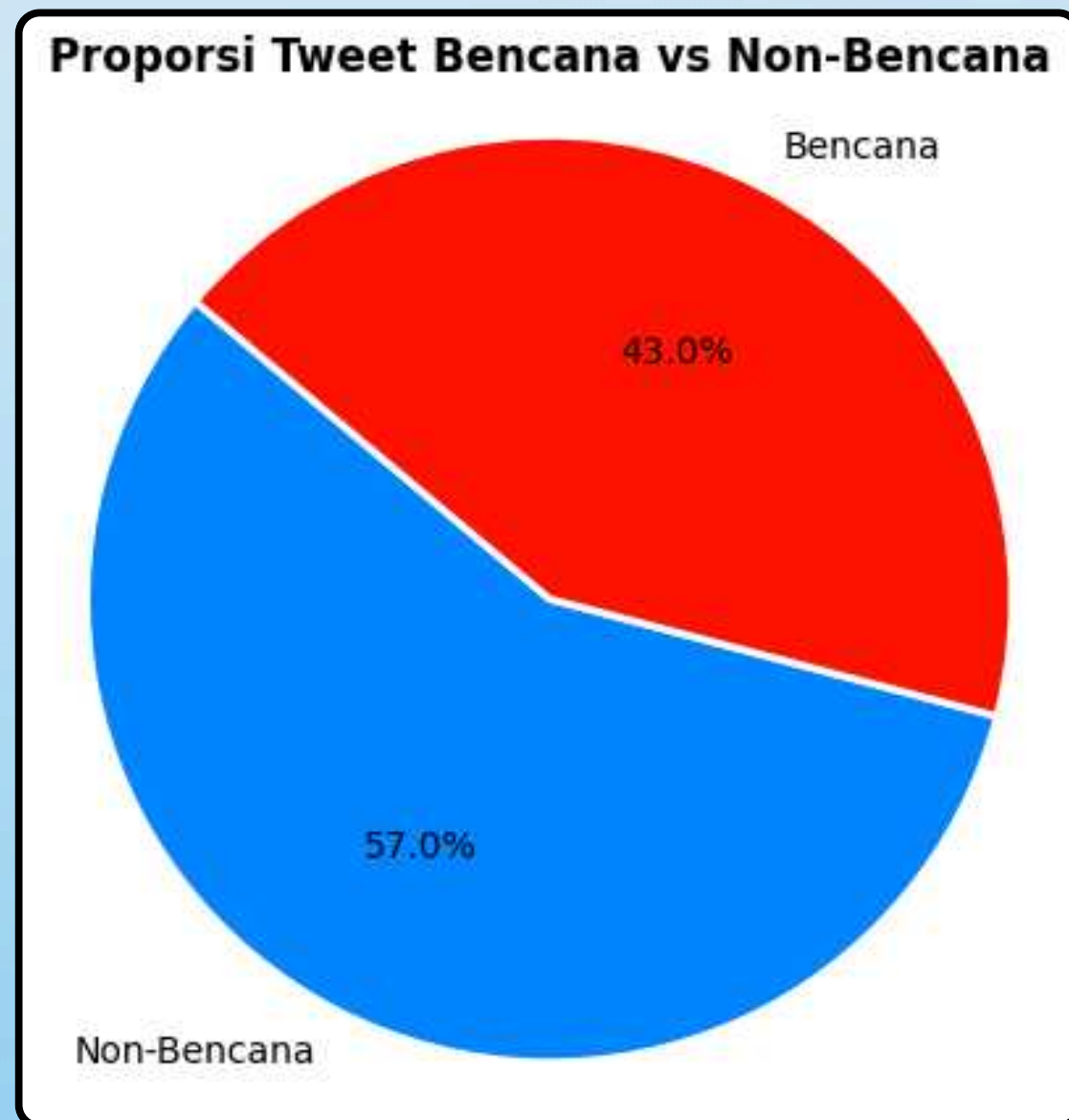




Exploratory Data Analysis



Proporsi Tweet

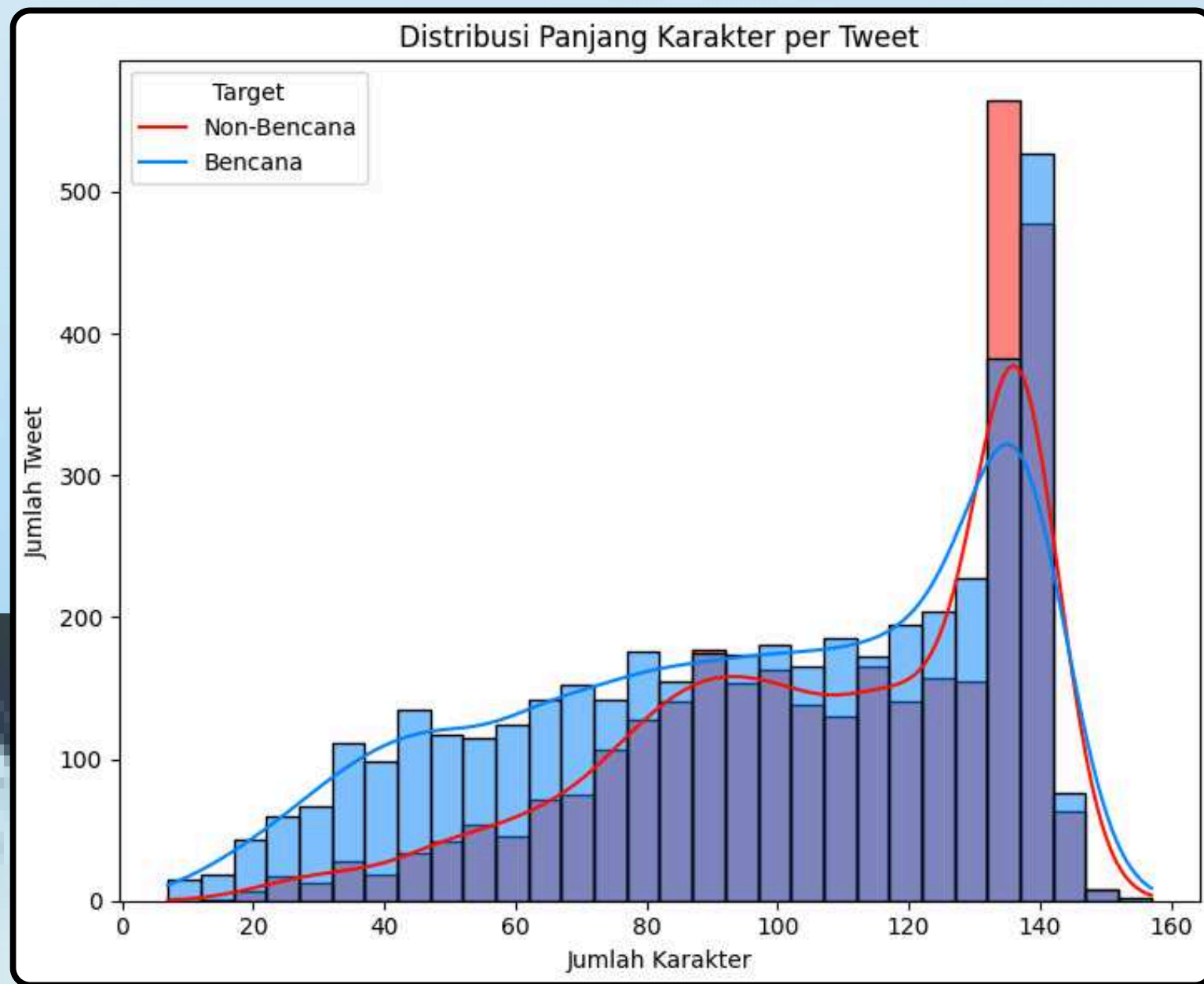


- Data relatif **seimbang**, sehingga model tidak perlu penanganan khusus untuk mengatasi imbalance data.
- Contoh Tweet Non-bencana:
 - "Burned my popcorn??"
 - "Sitting around a fire sounds great right about now"
- Contoh Tweet Bencana:
 - "Severe storm weakening as it moves SE towards Lubbock area. Outflow boundary may create dust and 50 mph gusts"
 - "Heavy Rainfall and Flooding in Northern #VietNam | Situation Report No.2"



Exploratory Data Analysis

Distribusi Panjang Karakter

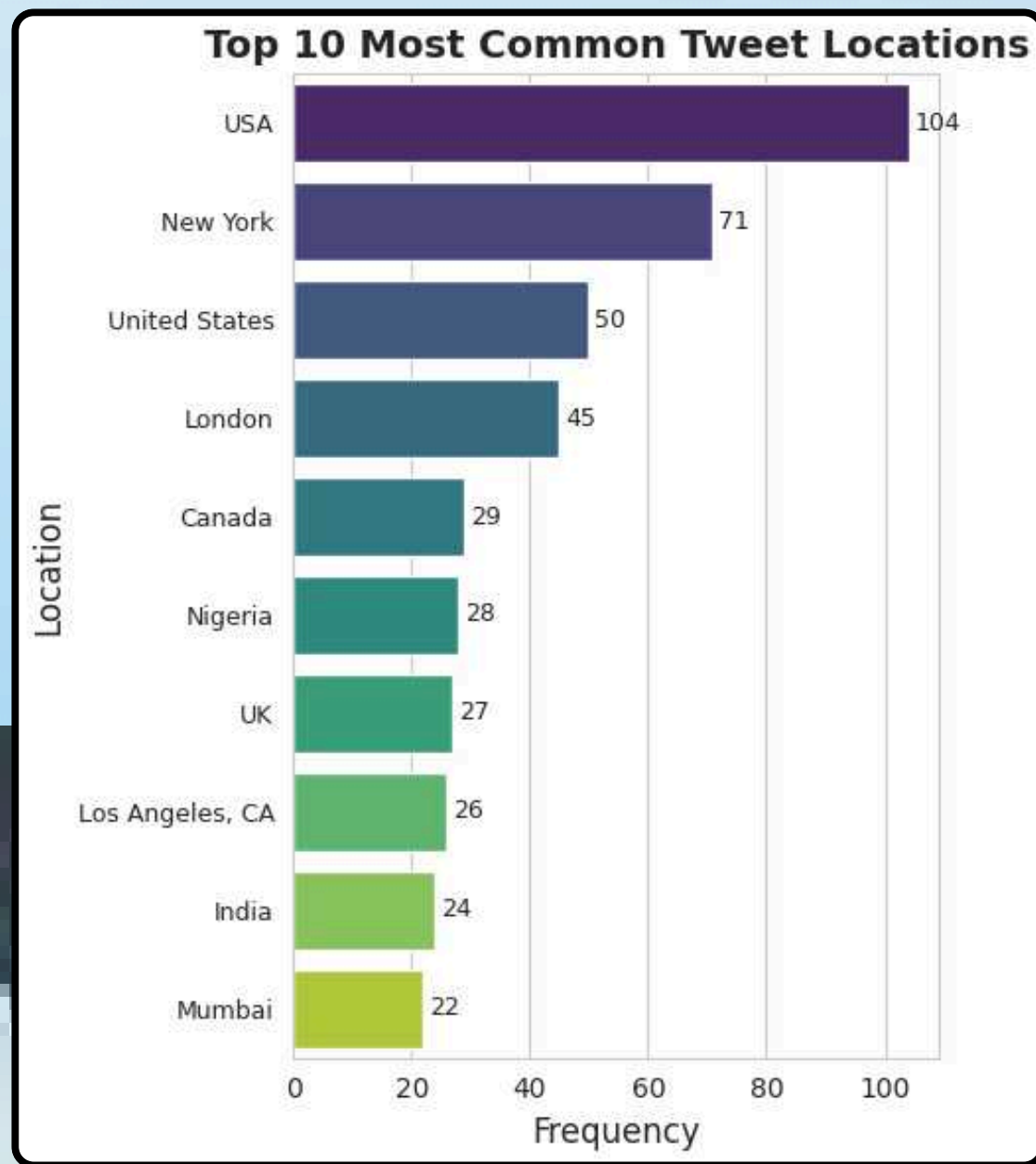


- Tweet tentang bencana cenderung lebih pendek dan langsung ke poin, sering muncul pada rentang 20-100 karakter. Sedangkan tweet non-bencana lebih sering memanfaatkan batas panjang maksimal.



Exploratory Data Analysis

Lokasi

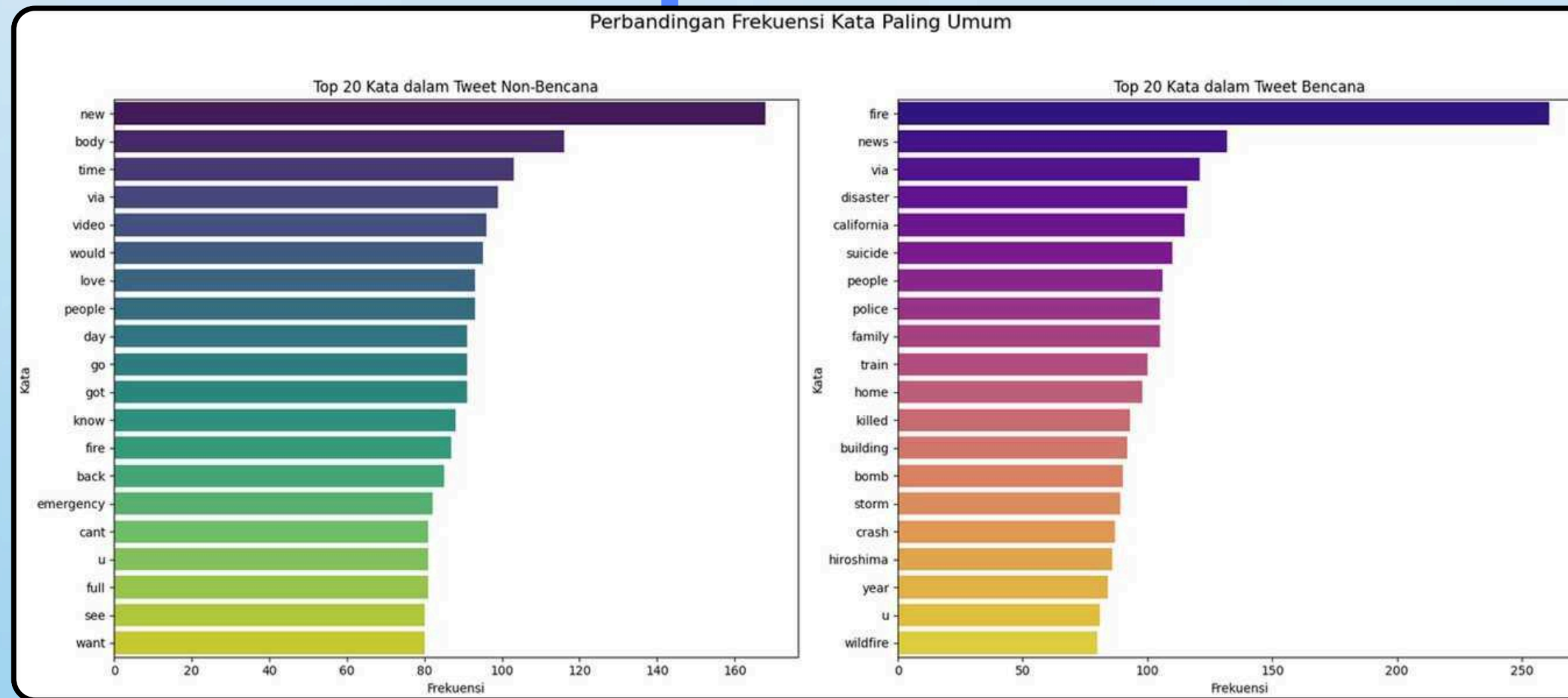


■ Lokasi yang ada di data ini hanya yang sudah diberi tahu sebelumnya. Masih banyak tweet yang belum diketahui lokasinya.



Exploratory Data Analysis

Top 20 word



Kosakata dalam tweet non-bencana bersifat umum, tidak spesifik, dan mencerminkan topik percakapan sehari-hari di media sosial.

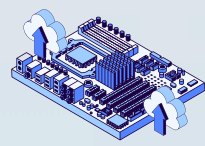
Kosakata dalam tweet bencana sangat bertema, spesifik, dan mengandung kata-kata dengan muatan makna negatif atau darurat yang kuat.



Data

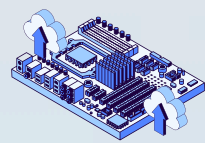
Preprocess





Langkah Data Preprocessing

- Mengubah Teks menjadi Huruf Kecil
- Mengekspansi contractions (I'm → I am)
- Menghapus URL
- Menghapus HTML Tag
- Menghapus Tanda Baca
- Menghapus Angka
- Tokenisasi
- Menghapus Stopwords dan Lemmatisasi
- Menghapus karakter non-ASCII & kata satu huruf



Contoh Tweet

Non-Bencana

- **Original** : @bbcmtd Wholesale Markets ablaze <http://t.co/IHYXEOHY6C>
- **Cleaned** : bbcmtd wholesale market ablaze

Bencana

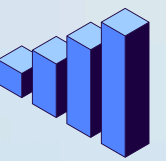
- **Original** : Deputies: Man shot before Brighton home set ablaze <http://t.co/gWNRhMSO8k>
- **Cleaned** : deputy man shot brighton home set ablaze



TF-IDF

Vectorization

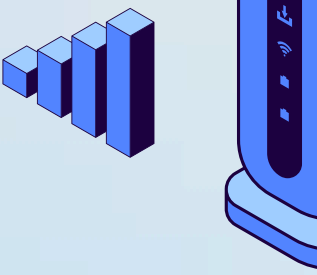




Text Vectorization (Logistic Regression)

■ Parameter TF-IDF terbaik:

- **max_df=0.75** : mengabaikan kata terlalu umum (muncul >75% dokumen)
- **min_df=1** : mempertahankan kata yg muncul minimal sekali
- **max_features=5000** : hanya ambil 5.000 kata/ngram terpenting
- **ngram_range=(1,2)** : gunakan unigram & bigram untuk tangkap konteks
- **stop_words='english'** : hapus stopword bahasa Inggris
- **sublinear_tf=False** : pakai tf normal tanpa log scaling



Text Vectorization (Naive Bayes)

■ Parameter TF-IDF terbaik:

- **max_df=0.9** : mengabaikan kata terlalu umum (>90% dokumen)
- **min_df=2** : hilangkan kata yg terlalu jarang (muncul <2 dokumen)
- **max_features=7000** : batasi pada 7.000 kata/ngram terpenting
- **ngram_range=(1,2)** : gunakan unigram & bigram untuk tangkap konteks
- **stop_words='english'** : hapus stopword bahasa Inggris
- **sublinear_tf=True** : gunakan log-scaling untuk menekan pengaruh kata sangat sering

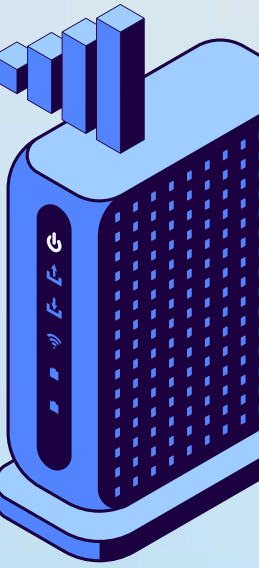


Features Selection





Features Selection (SelectKBest)



- Menggunakan uji χ^2 (chi-square) untuk memilih 5.000 kata/ngram yang paling relevan dengan target (bencana vs non-bencana).

- Membantu mempercepat training, mengurangi overfitting, dan meningkatkan performa model.





Split Data



- Data dibagi 80% training – 20% testing.
- Stratified → proporsi Non-Bencana & Bencana tetap seimbang.
- Random seed → hasil selalu konsisten & bisa direplikasi.





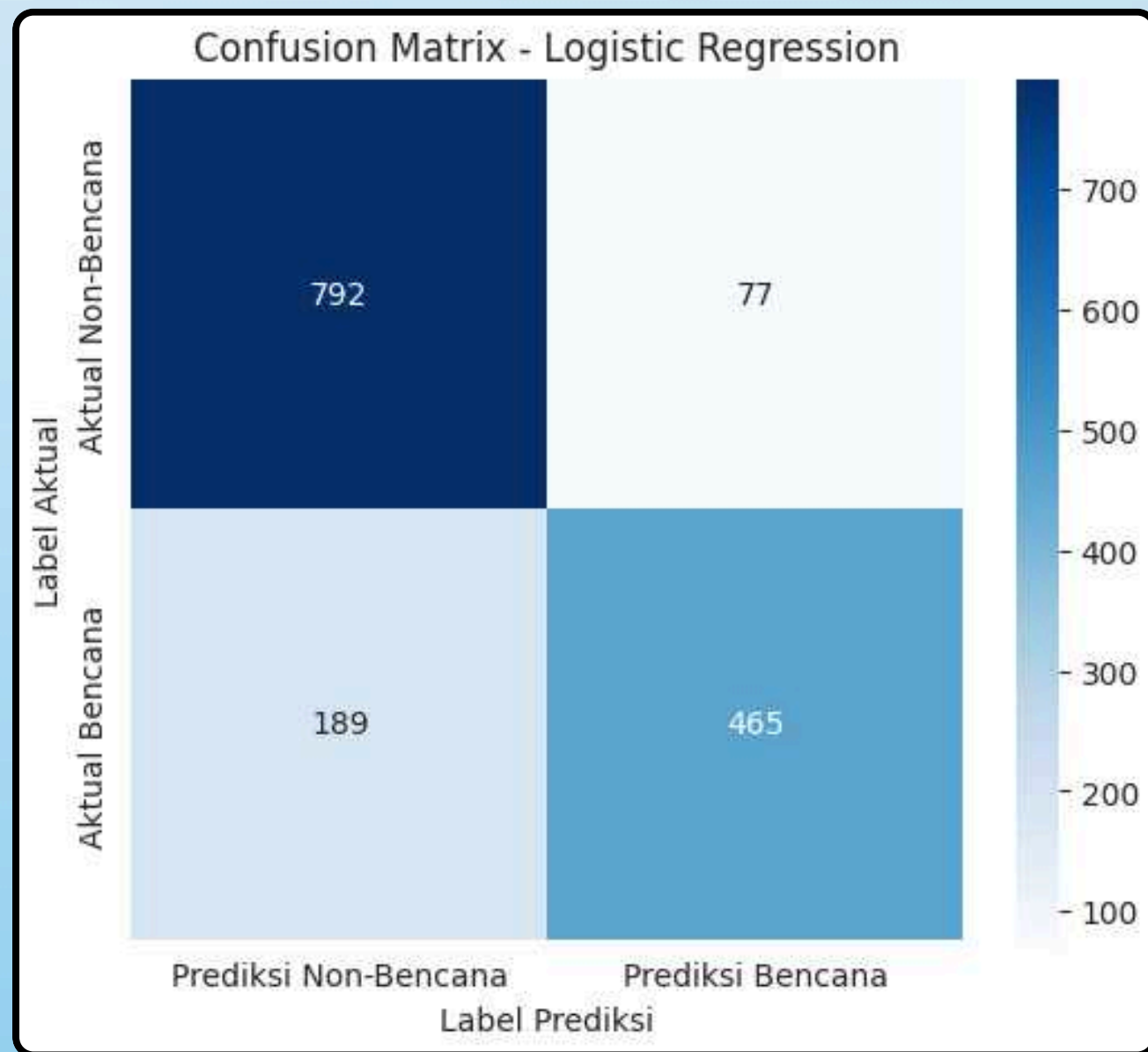
Logistic

Regression

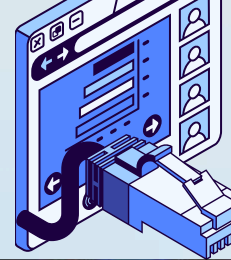




Evaluasi Base Model Logistic Regression

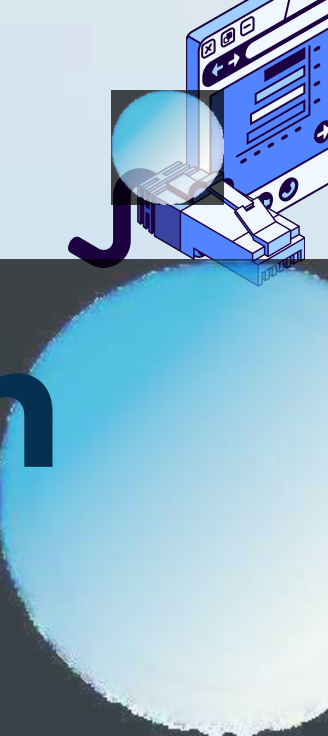


	Kategori 0 (Non-Bencana)	Kategori 1 (Bencana)
precision	0.81	0.86
recall	0.91	0.71
f1-score	0.86	0.78

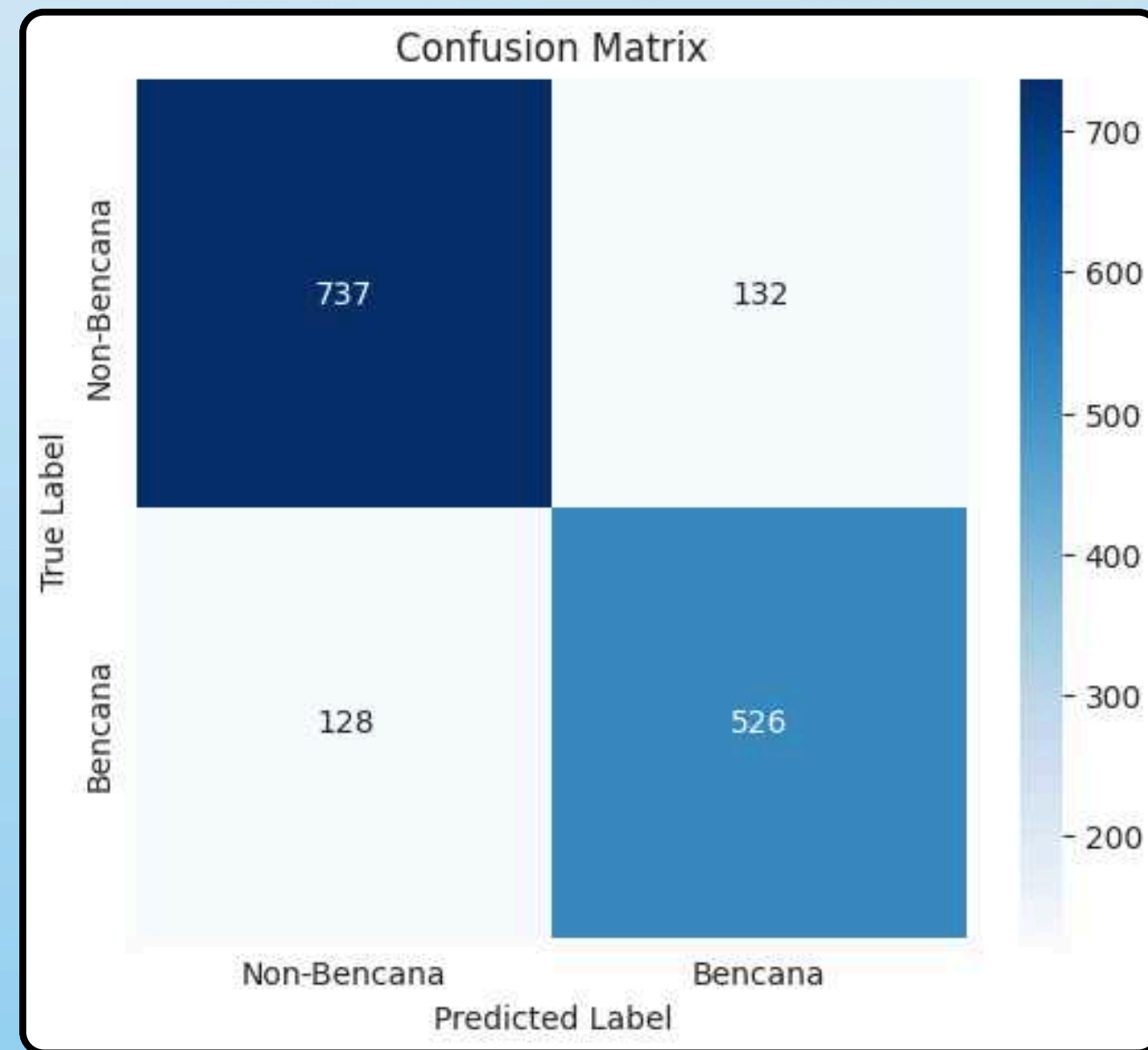


Hyperparameter Tuning Logistic Regression

Parameter	Value yang diuji	Penjelasan
C	[0.01, 0.1, 1, 10, 50]	Kontrol kekuatan regularisasi (semakin kecil, semakin kuat regularisasi).
penalty	l2	Jenis regularisasi. l2 → menambahkan penalti kuadrat bobot agar model tidak overfit.
solver	liblinear	Algoritma optimasi untuk mencari bobot terbaik.
class_weight	[None, balanced]	Penyeimbang bobot kelas (balanced untuk dataset imbalance).
max_iter	[100, 200, 500]	Batas maksimum iterasi solver dalam mencari bobot.
scoring	f1_score (kelas 1)	Fokus F1 kelas 1 (bencana).



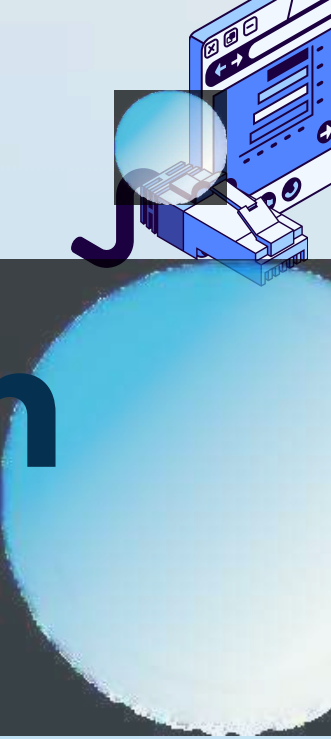
Evaluasi Best Model Logistic Regression



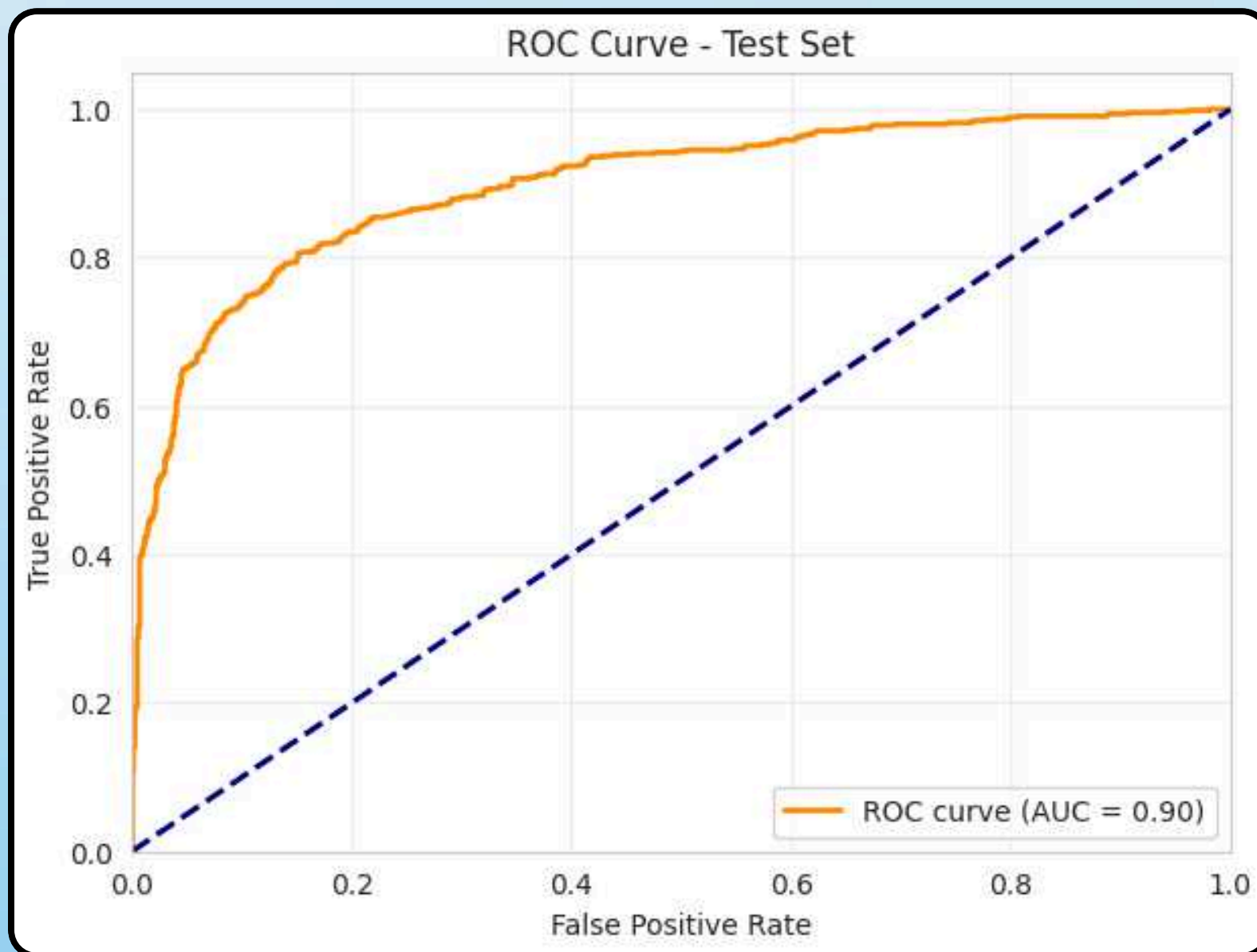
Parameter:

- C: 10,
- class_weight: balanced,
- max_iter: 100,
- penalty: l2,
- solver: liblinear

	Kategori 0 (Non-Bencana)	Kategori 1 (Bencana)
precision	0.85	0.80
recall	0.85	0.80
f1-score	0.85	0.80



Evaluasi Best Model Logistic Regression

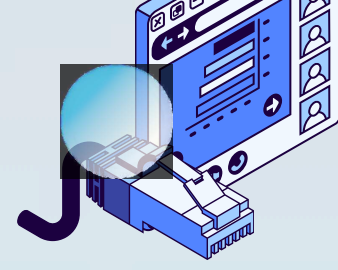


- **AUC 0.90:** model memiliki kemampuan yang sangat baik dalam membedakan antara kelas positif dan negatif.



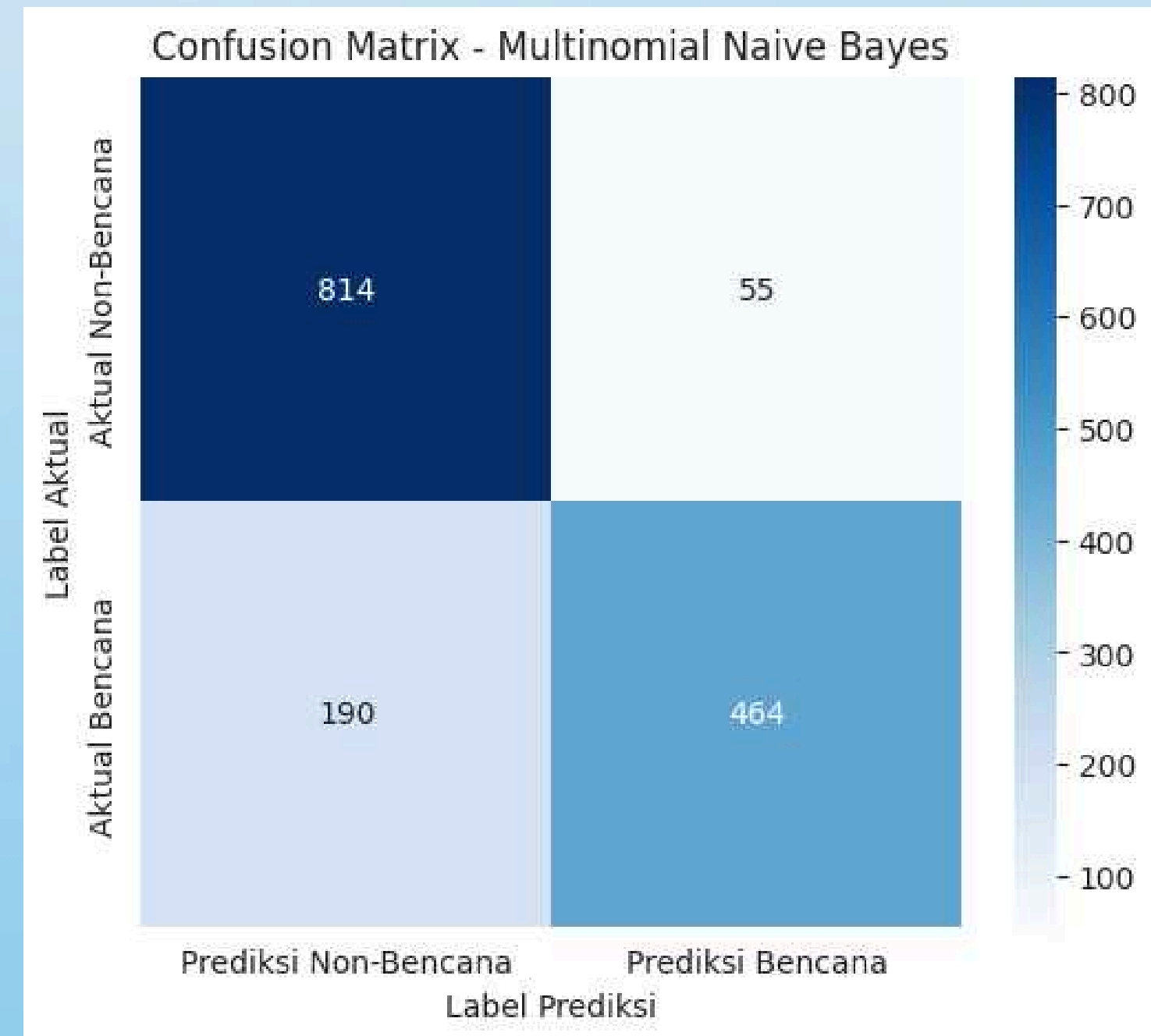
Naive Bayes

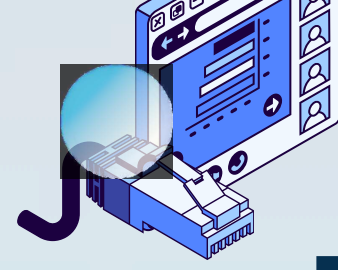




Evaluasi Base Models Naive Bayes

	Kategori 0 (Non-Bencana)	Kategori 1 (Bencana)
precision	0.81	0.89
recall	0.94	0.71
f1-score	0.87	0.79

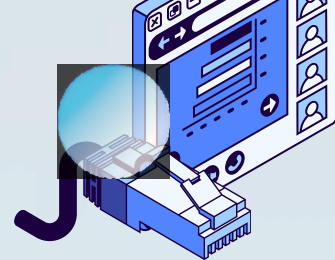




Hyperparameter Tuning Naive Bayes

GridSearchCV with 5-fold cross-validation

Parameter	Value yang diuji	Penjelasan
alpha	[0.001, 0.01, 0.1, 0.5, 1.0]	Smoothing untuk menangani kata-kata baru agar tidak menghasilkan probabilitas nol.
fit_prior	[True, False]	Menentukan apakah probabilitas awal kelas akan dipelajari dari data (True) atau diasumsikan seragam/sama (False).
class_prior	[[0.4, 0.6], [0.45, 0.55], [0.5, 0.5]] (hanya diuji jika fit_prior=True)	Menetapkan probabilitas awal kelas secara manual, menguji berbagai asumsi distribusi. Hanya berjalan jika fit_prior=True.

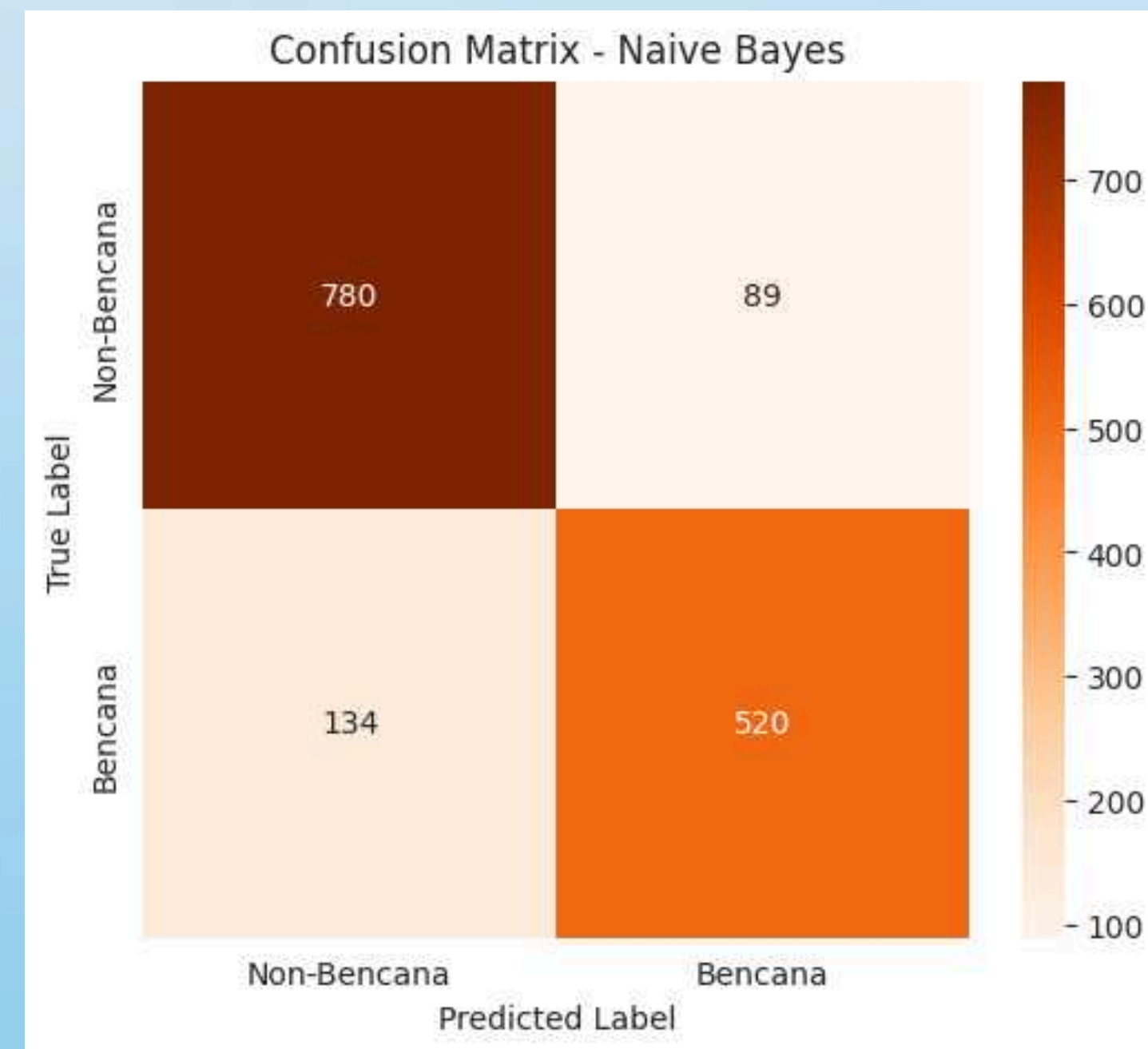


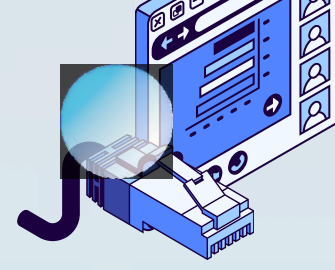
Evaluasi Best Models Naive Bayes

Parameter:

- alpha : 0.01
- class_prior: [0.5, 0.5],
- fit_prior = True

	Kategori 0 (Non-Bencana)	Kategori 1 (Bencana)
precision	0.85	0.85
recall	0.90	0.80
f1-score	0.87	0.82





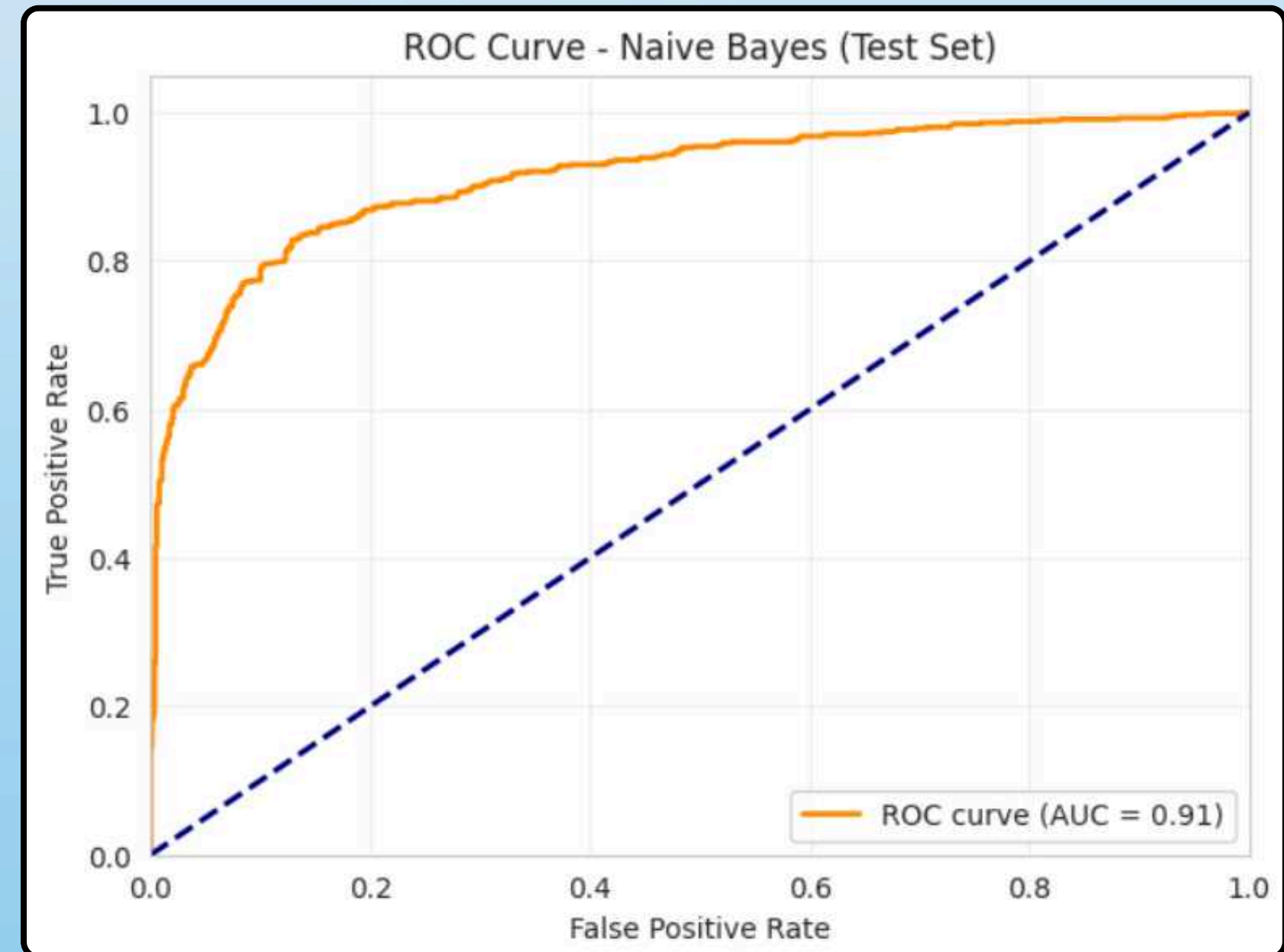
Evaluasi Best Models Naive Bayes

Kurva ROC (garis orange) melengkung jauh ke pojok kiri atas menandakan performa yang sangat baik. Artinya, model mampu mencapai True Positive Rate yang tinggi dengan False Positive Rate yang rendah.

AUC (Area Under the Curve)

Skor 0.91 menunjukkan bahwa model memiliki kemampuan yang sangat baik dalam memisahkan kedua kelas.

Jarak yang signifikan antara kurva ROC dan garis biru ini mengindikasikan bahwa model Naive Bayes sangat baik dalam memprediksi.





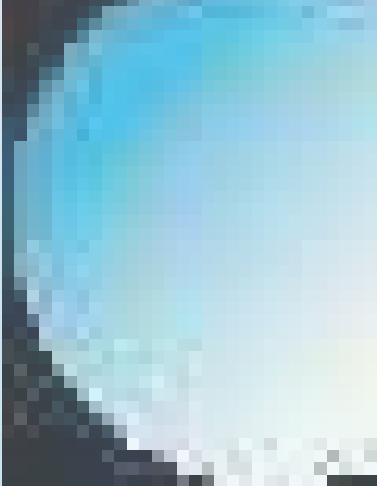
Model

Interpretability





Hasil Model Interpretability



text	actual_label	predicted_label_lr	predicted_label_nb
Haley Lu Richardson Fights for Water in The La...	0	1	1
Jack Wilshere has poor injury recordand his of...	1	1	0
#360WiseNews : China's Stock Market Crash: Are...	1	0	0
Path of Obliteration\nBack From The Dead\nStor...	1	0	0
New post from @darkreading http://t.co/8eIJDXA...	1	0	0
@leedsrouge Love what you picked! We're playin...	1	0	0
'i never understood guys who wanted to run int...	0	1	0
From recycling to only using non-hazardous che...	1	0	1
Is This Country Latin America's Next 'Argentin...	0	1	1
#LakeEffect #Snowstorm Twill Denim Jackets *...	1	0	0



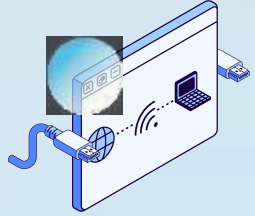
Kata Ambigu Diprediksi Model

text	actual_label	predicted_label_lr	predicted_label_nb
Crying out for more! Set me ablaze	0	1	1
on the outside you're ablaze and alive but you're dead inside	0	0	1
TRUCK ABLAZE : R21. VOORTREKKER AVE. OUTSIDE OR TAMBO INTL. CARGO SECTION. http://t.co/8kscqKfKkF	1	1	0
Progressive greetings! In about a month students would have set their pens ablaze in The Torch Publications'... http://t.co/9FxFiXQuJt	0	0	1

■ Contoh kesalahan prediksi model pada tweet yang mengandung kata 'ablaze'. Baris-baris ini menunjukkan perbedaan antara label aktual dan hasil prediksi model LR atau NB.

■ Masih terdapat beberapa kesalahan prediksi termasuk untuk tweet yang mengandung kata 'ablaze'.

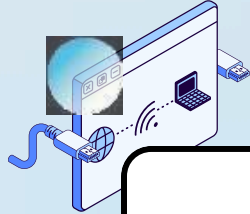
- LR hanya **satu kesalahan** prediksi
- NB ada **4 kesalahan** prediksi



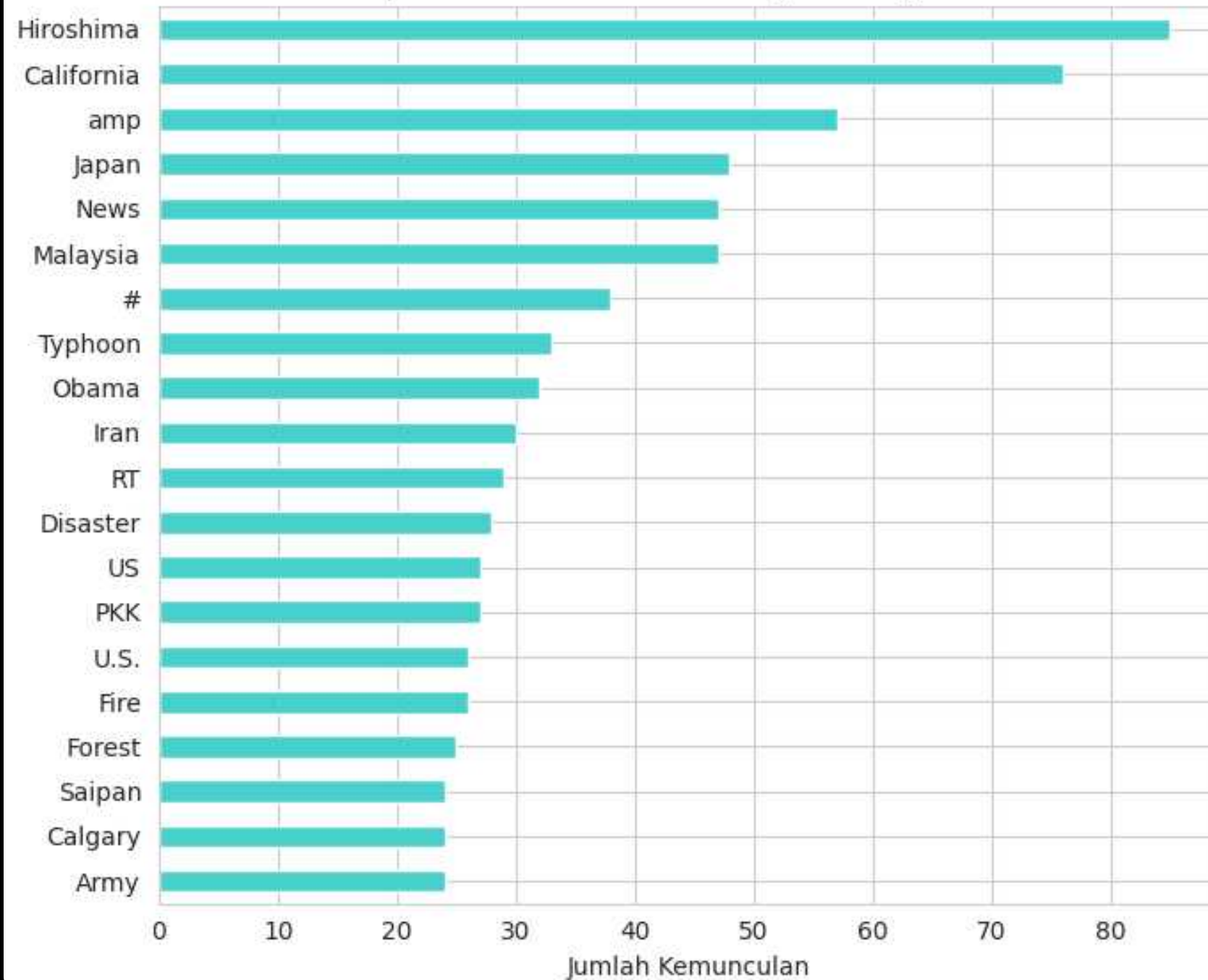
Mengekstrak Lokasi

text	extracted_location_lr	extracted_location_nb
#world FedEx no longer to transport bioterror germs in wake of anthrax lab mishaps http://t.co/5zDbTktwW7	FedEx	FedEx
How One Couple Is Using Drones to Save Refugees at the World's Deadliest Border http://t.co/9qpG0Z3Rh9	Border, Couple, Deadliest, Drones, Refugees, World	Border, Couple, Deadliest, Drones, Refugees, World
they say bad things happen for a reason but no wise words gonna stop te bleeding	Unknown	Unknown
Help me survive the zombie apocalypse at the Zombie Fun Run on November 15th. https://t.co/kgSwhSr7Mn #teamsurvivors #zombiefunrun2014	Unknown	Unknown
Former Freedom Surf/Spa Razed to Make Way for New Homes http://t.co/EX6JzQJ3NI	Unknown	Unknown
Experts in France begin examining airplane debris found on Reunion Island http://t.co/LsMx2vwr3J French air accident experts on Wednesday□Ų_	France, Reunion Island	France, Reunion Island
I have the biggest crush on you & I dont know if you'll ever know it ??	Unknown	Unknown
Wow what beauty lies within every storm. Taken today after a rainstorm at A&B Pipeliners http://t.co/pSt5bBQ0av	A&B, Pipeliners	A&B, Pipeliners
8/6/2015@2:09 PM: TRAFFIC ACCIDENT NO INJURY at 2781 WILLIS FOREMAN RD http://t.co/VCKIT6EDEv	FOREMAN, RD, WILLIS, http://t.co/VCKIT6EDEv	FOREMAN, RD, WILLIS, http://t.co/VCKIT6EDEv
It's like God wants me to become a mass murderer with how many dickheads I have to deal with on a daily basis.	Unknown	Unknown

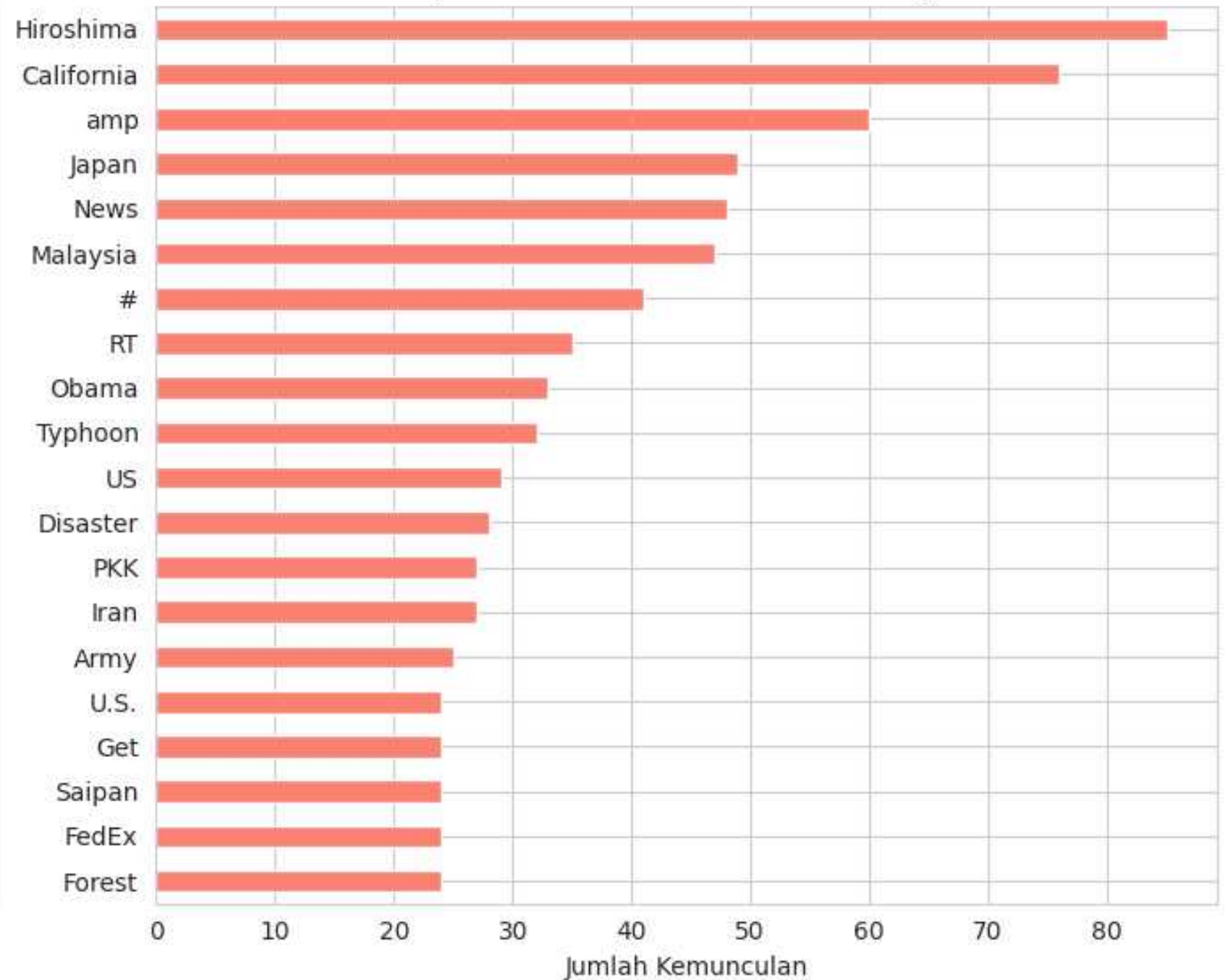
Mengekstrak Lokasi



Top 20 Lokasi Terdeteksi - Logistic Regression

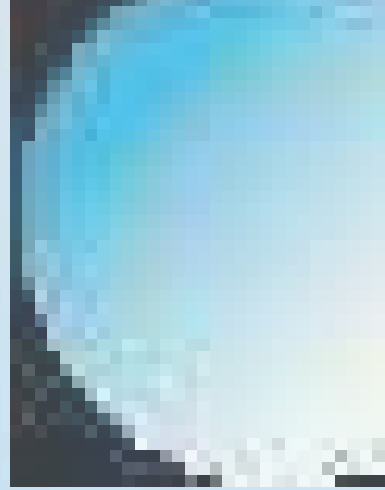


Top 20 Lokasi Terdeteksi - Naive Bayes

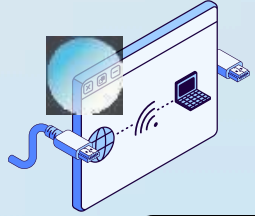




Mendeteksi EVENT

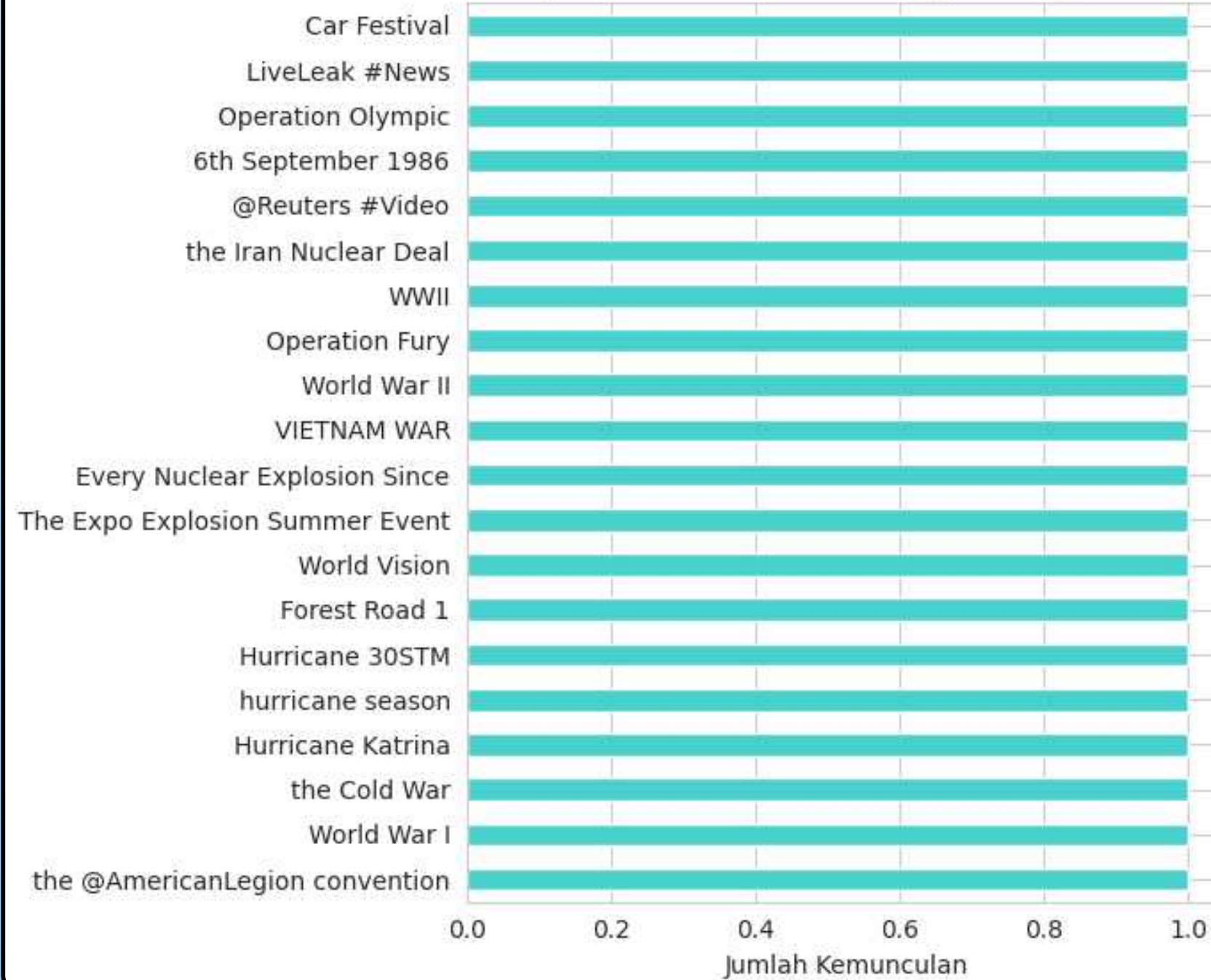


text	extracted_event_time_lr	extracted_event_time_nb
You Are Invited to The Expo Explosion Summer Event 2015! WHEN: August 14th Friday 2015 WHERE: Ben E Keith... http://t.co/yh4R7Ug21a	EVENT: The Expo Explosion Summer Event DATE: August 14th Friday 2015	EVENT: The Expo Explosion Summer Event DATE: August 14th Friday 2015
Ukraine argues for US weapons during Pelosi's visit to Kiev http://t.co/jnN0kRNxvY http://t.co/5LOiWuyv5r	EVENT: Pelosi	EVENT: Pelosi
Wow-the name #LegionnairesDisease comes from an outbreak of pneumonia at the @AmericanLegion convention in Philly in 1976--29 died from it.	EVENT: the @AmericanLegion convention DATE: 1976--29	EVENT: the @AmericanLegion convention DATE: 1976--29
#Laos flooding -10 villages underwater World Vision responding http://t.co/8fvQRizOux	EVENT: World Vision	EVENT: World Vision
A GPM satellite 'bullseye' in Typhoon Soudelor http://t.co/7vcEzi6CbB	EVENT: Typhoon Soudelor http://t.co/7vcEzi6CbB	EVENT: Typhoon Soudelor http://t.co/7vcEzi6CbB
DUST IN THE WIND: @82ndABNDIV paratroopers move to a loading zone during a dust storm in support of Operation Fury: http://t.co/uGesKLCn8M	EVENT: Operation Fury	EVENT: Operation Fury
A Time-Lapse Map of Every Nuclear Explosion Since 1945 - by Isao Hashimoto #War #atomicbomb https://t.co/V0t8H4lecc	EVENT: Every Nuclear Explosion Since DATE: 1945	EVENT: Every Nuclear Explosion Since DATE: 1945
Debris confirmed from MH370; relatives hope for discovery of crash site http://t.co/rLFtjmHHvT via @Reuters #Video	EVENT: @Reuters #Video	EVENT: @Reuters #Video
@BldrCOSheriff says 2nd grenade found is 'younger' than WWII era grenade found earlier. They'll detonate it at 8:00 tonight. @CBSDenver	EVENT: WWII TIME: 8:00 tonight	EVENT: WWII TIME: 8:00 tonight
the_af comments on 'New evidence of Japan's effort to build atom bomb at the end of WWII' - http://t.co/hcTxghR2Yf	Unknown	EVENT: WWII

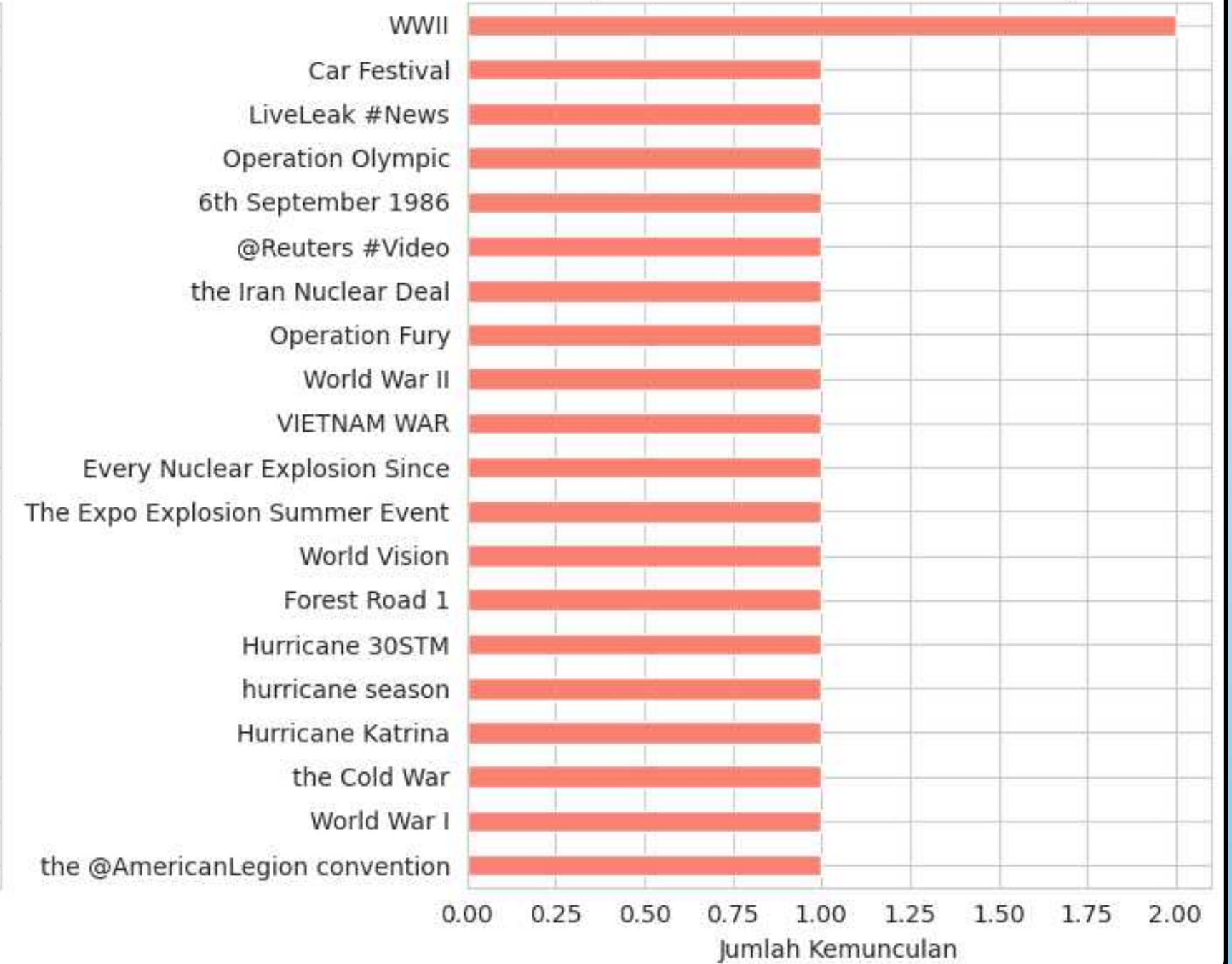


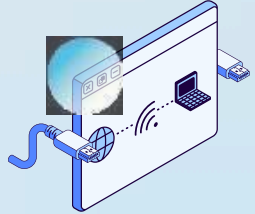
Mendeteksi EVENT

Top 20 Event Terdeteksi - Logistic Regression



Top 20 Event Terdeteksi - Naive Bayes





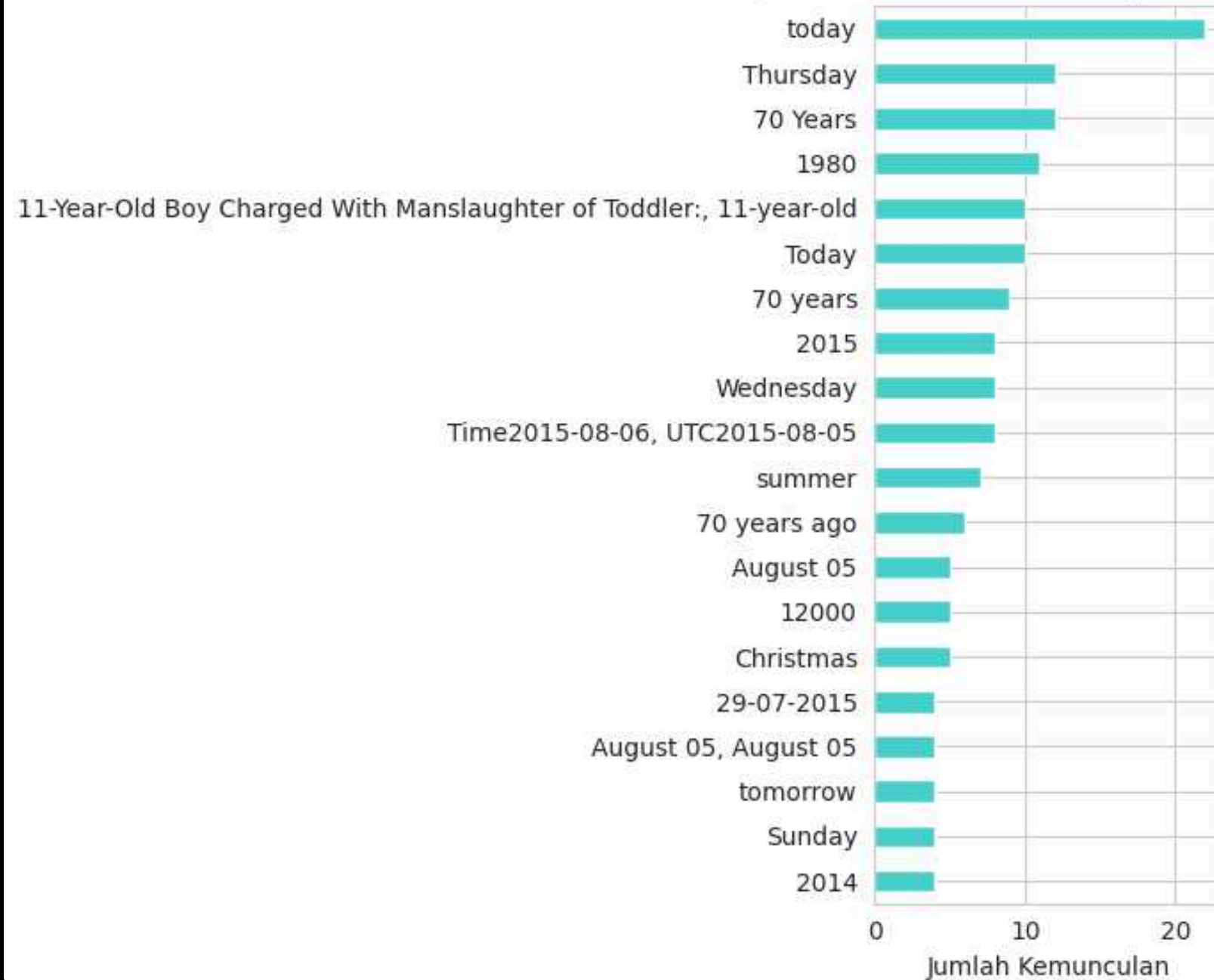
Mendeteksi DATE

text	extracted_event_time_lr	extracted_event_time_nb
I'm grateful for our first responders - @MNPDNashville @NashvilleFD and others - who acted so swiftly to save countless lives today.	DATE: today	DATE: today
□Ū@SplottDave: @TeamPalestina That's about 28700 Palestinian homes demolished now by Israel since 1967 w/ 0 Israeli home demolished @POTUS	DATE: 1967	DATE: 1967
You Are Invited to The Expo Explosion Summer Event 2015! WHEN: August 14th Friday 2015 WHERE: Ben E Keith... http://t.co/yh4R7Ug21a	EVENT: The Expo Explosion Summer Event DATE: August 14th Friday 2015	EVENT: The Expo Explosion Summer Event DATE: August 14th Friday 2015
600 passengers abandoned at LRT station during Tuesday's hailstorm http://t.co/vgF41luPkn #yyc #yycstorm #abstorm	DATE: Tuesday	DATE: Tuesday
#USGS M 0.9 - Northern California: Time2015-08-06 01:50:25 UTC2015-08-06 01:50:25 UTC at epicenter... http://t.co/mBo6OAnlQI #EarthTwerk	DATE: Time2015-08-06, UTC2015-08-06	DATE: Time2015-08-06, UTC2015-08-06
wowo--=== 12000 Nigerian refugees repatriated from Cameroon	DATE: 12000	DATE: 12000
It's been 70 years (and one hour) since the bombing of Hiroshima. Let's take this time to remember.	DATE: 70 years TIME: one hour	DATE: 70 years TIME: one hour
#USGS M 1.9 - 5km S of Volcano Hawaii: Time2015-08-06 01:04:01 UTC2015-08-05 15:04:01 -10:00 at epicenter... http://t.co/dlsrwhQGym #SM	DATE: Time2015-08-06, UTC2015-08-05	DATE: Time2015-08-06, UTC2015-08-05
.@StacDemon with five burning questions for Chris Mullin and St. John□Ū's in 2015-16: http://t.co/NmRVTHkvAh #SJUBB	DATE: 2015-16	DATE: 2015-16
I moved to England five years ago today. What a whirlwind of time it has been! http://t.co/eaSIGeA1B7	DATE: five years ago	DATE: five years ago



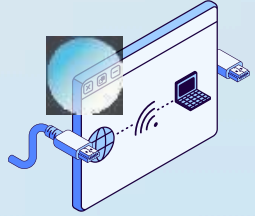
Mendeteksi DATE

Top 20 DATE Terdeteksi - Logistic Regression

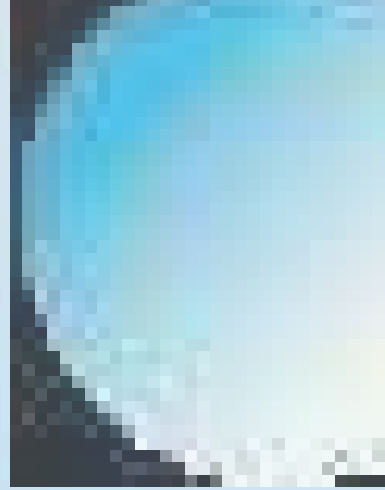


Top 20 DATE Terdeteksi - Naive Bayes





Mendeteksi TIME

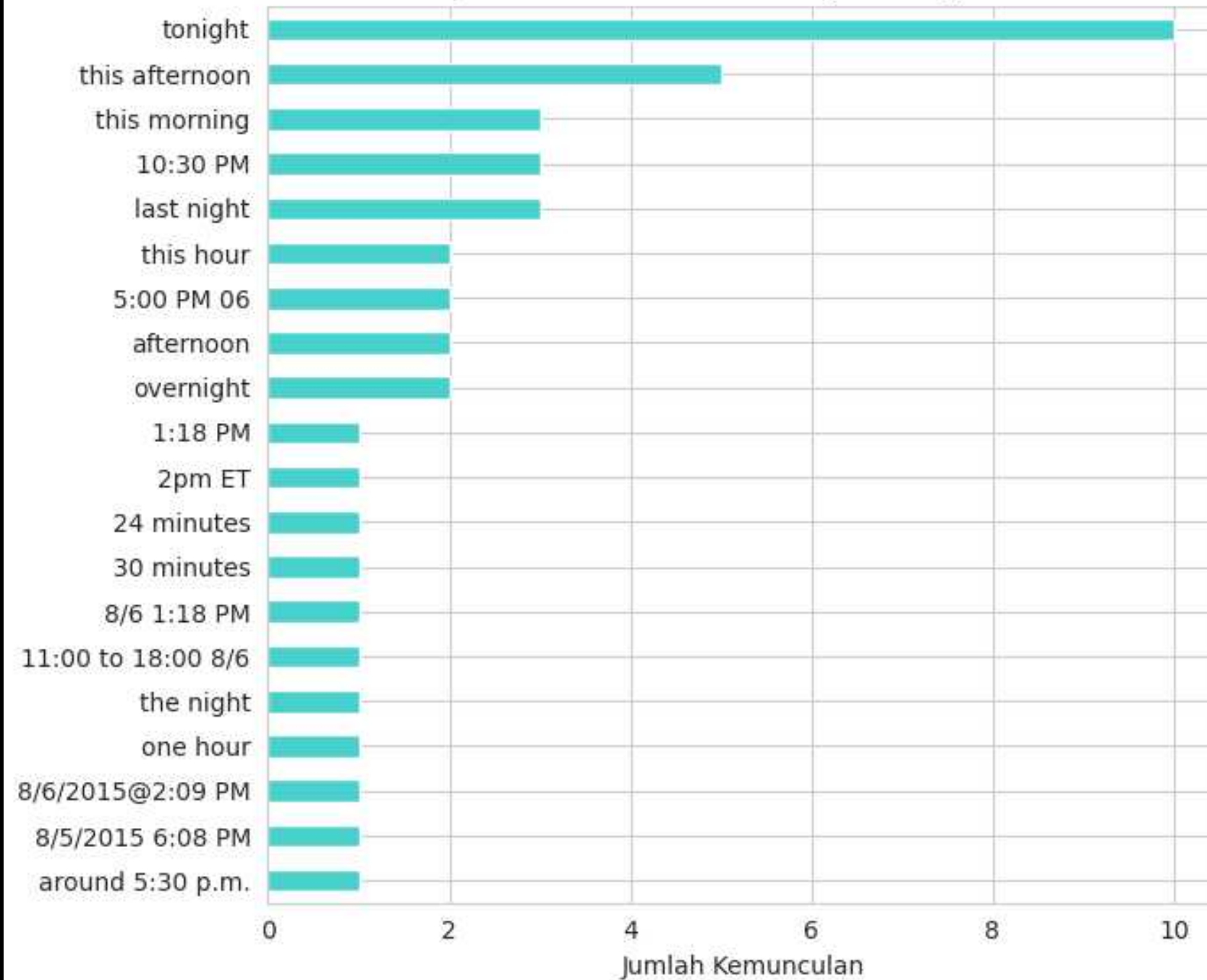


text	extracted_event_time_lr	extracted_event_time_nb
@MissJadeBrown tells of the tragic mid-morning collision which claimed the life of a young man riding a motorcycle. https://t.co/rPDA60Aoni	TIME: mid-morning	TIME: mid-morning
My baby girls car wreck this afternoon thank God no serious injuries and she was wearing her seatbelt!!!!... http://t.co/NJQV45ndS2	TIME: this afternoon	Unknown
When there's a fire alarm going off in zehrs and we keep working for 20 minutes then decide to evacuate everyone..	TIME: 20 minutes	TIME: 20 minutes
NWS says thunderstorms with deadly lightning will move across the Black Hills this evening. That assumes there's a safe type. Hmm?	Unknown	TIME: this evening
@gilmanrocks7 wow. Where is this? For the rest of the summer I usually have to worry about forest fires. Calgary is about 8 hours away.	TIME: about 8 hours	TIME: about 8 hours
Couple storms near Guthrie OK. Leaving Norman for an evening lightning op. #okwx http://t.co/HcwrK81p71	TIME: evening	TIME: evening
oh yeah my ipod almost exploded last night i was using it while charging and shit was sparking akxbskdn almost died	TIME: last night	TIME: last night
8/6/2015@2:09 PM: TRAFFIC ACCIDENT NO INJURY at 2781 WILLIS FOREMAN RD http://t.co/VCKIT6EDEv	DATE: 2781 TIME: 8/6/2015@2:09 PM	DATE: 2781 TIME: 8/6/2015@2:09 PM
@OhYayyyYay the train derailed this morning	TIME: this morning	TIME: this morning
Beat:B2 MOTOR VEHICLE COLLISION at N 35 ST / FREMONT AV N reported on 8/5/2015 6:52 PM Call# 15000270364	TIME: 8/5/2015 6:52 PM	TIME: 8/5/2015 6:52 PM

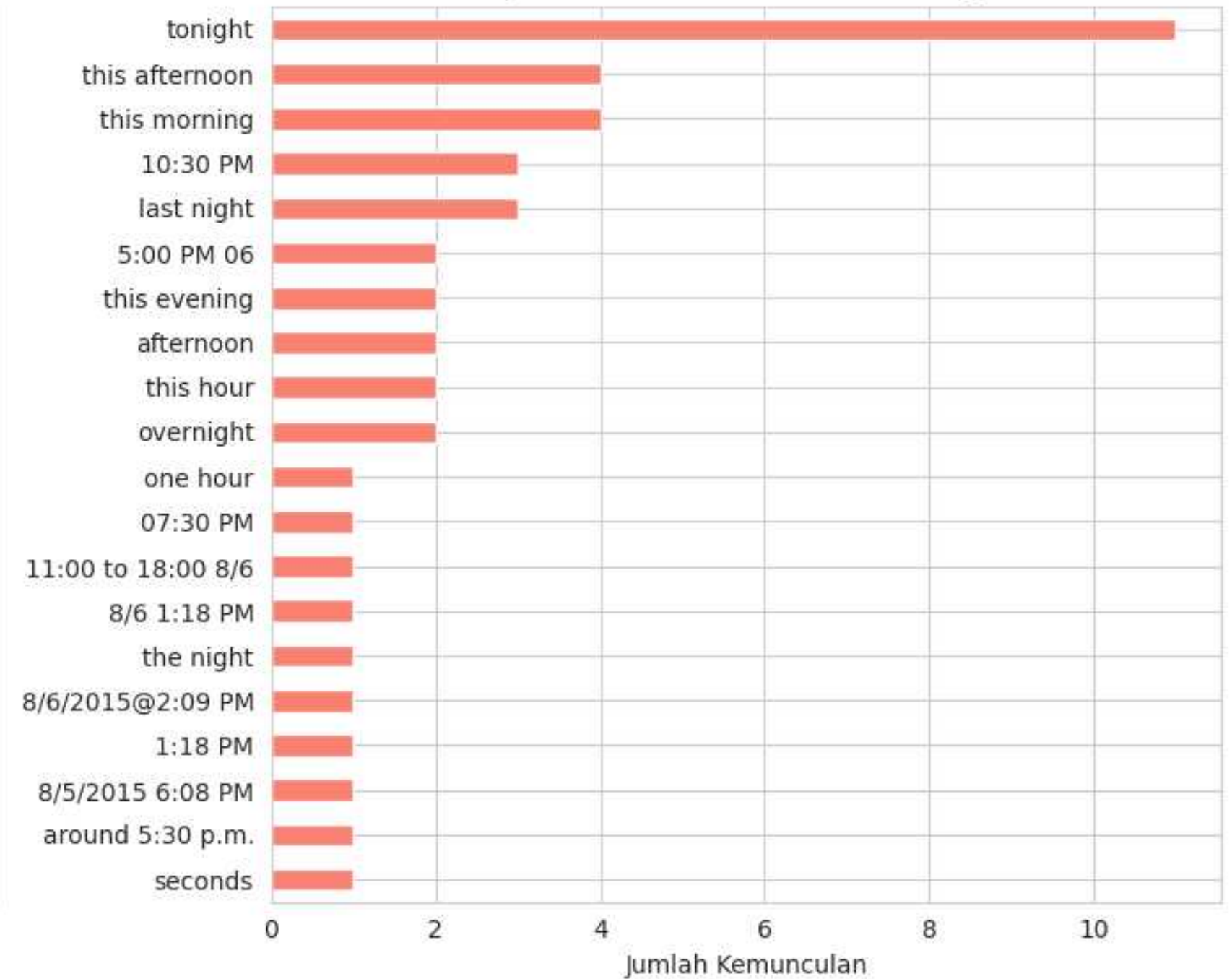


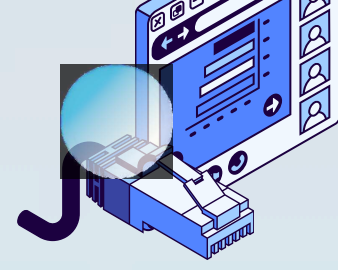
Mendeteksi TIME

Top 20 TIME Terdeteksi - Logistic Regression



Top 20 TIME Terdeteksi - Naive Bayes





Insight

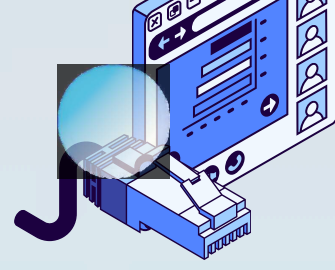


- Kedua model sudah termasuk sangat baik dalam memprediksi. Naive Bayes sedikit lebih unggul dibandingkan dengan Logistic Regression

- Lokasi yang terdeteksi sering ambigu atau muncul karena konteks tidak relevan.

- Contoh pada kata "ablaze" menunjukkan bahwa model yang sudah dioptimalkan pun masih bisa salah. Model kesulitan membedakan beberapa penggunaan metaforis seperti "set me ablaze".

- Mendeteksi event, date, time juga masih terdapat yang kurang tepat.



Kesimpulan

- Fokus pada recall dan f1-score di kelas Bencana, model Logistic Regression dan Naive Bayes merupakan model yang baik untuk kasus klasifikasi data text
- Kata-kata ambigu seperti “ablaze” dapat menyebabkan kebingungan klasifikasi, sehingga konteks kalimat harus diperhatikan dalam train model.
- Model interpretability menggunakan Named Entity Recognition (NER) dapat digunakan untuk mengekstraksi informasi lokasi dan waktu kejadian bencana, meskipun masih terbatas karena tidak semua tweet memuat informasi tersebut secara eksplisit.





Thank You!

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With us**

